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BCA (Sem-III) – Computer

Network (303) Core-VII

2025

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer all group as directed.

Group—A

(Objective Type Questions)

1. Choose the correct answer : $2 \times 10 = 20$

(i) Which ARQ protocol retransmits all frames after an error occurs ?

A) Stop-and-Wait ARQ

B) Selective Repeat ARQ

C) Go-Back-N ARQ

D) Sliding Window ARQ

(ii) Which of the following is a guided transmission medium?

A) Radio waves

B) Microwave

C) Optical Fiber Cable

D) Infrared

(iii) In which switching technique is a dedicated communication path established?

A) Packet Switching

B) Message Switching

C) Circuit Switching

D) Datagram Switching

(iv) In Frequency Division Multiplexing (FDM), signals are separated by:

A) Time slots

B) Different frequencies

C) Different IP addresses

D) Packet size

(v) Which handshake is generally used for TCP connection release?

A) Two-Way Handshake

B) Three-Way Handshake

C) Four-Way Handshake

D) One-Way Handshake

(vi) CSMA/CD is mainly used in:

A) Wireless Networks

B) Ethernet (Wired Networks)

C) Satellite Communication

D) Bluetooth

(vii) Which of the following is an example of a domain name?

A) 192.168.1.1

B) www.google.com

C) 255.255.255.0

D) MAC Address

(viii) What is the main function of DNS?

A) Encrypt data

B) Convert domain names into IP addresses

C) Compress files

D) Send emails

(ix) What is the main purpose of a routing table?

A) To store passwords

B) To determine the best path for packet forwarding

C) To store user data

D) To increase bandwidth

(x) In the Distance Vector Algorithm, routers share information with:

A) All routers in the internet

B) Only neighboring routers

C) End users

D) Servers only

Group – B

(Short Answer Type Questions)

Answer any **four** of the following : 5×4=20

2. Explain Transport Layer of OSI reference model.

3. Differentiate between guided and unguided transmission media with suitable examples.

4. What happens after collision occurs in CSMA/CD?

5. What is Hybrid Topology? Mention its advantages and disadvantages.

6. Differentiate between Circuit Switching and Packet Switching.

7. Write the OSI layer names at which the following network devices operate: Repeater, Hub, Switch, Bridge, and Router.

Group – C

(Long Answer Type Questions)

Answer any **two** of the following : 15×2=30

8. Explain the process of connection establishment and release in TCP using the Three-Way Handshake mechanism.

9. Discuss the disadvantages/problems of the Stop-and-Wait protocol. How are these problems resolved using the Stop-and-Wait ARQ protocol?

10. Explain the working principle of TDM in detail with a suitable diagram.

11. A message 1101011001 is to be transmitted using the CRC (Cyclic Redundancy Check) method with the generating polynomial $x^3 + x^2 + 1$ for error detection. Determine the transmitted message and explain the error-checking process at the receiver side.

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